

RaceDay Event Management System

Programming6212 Part 1: System Planning

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SECTION A: Entity Relationship Diagram

RaceDay is a web-based, end-to-end event management system developed in South Africa specifically for the road running, walking and cycling community. The platform provides event organizers with the ability to set up a event, with multiple categories and participants' results, and to manage these, and lets participants look for events, register to events and follow their performances.

This design document is primarily intended to specify the database structure and RESTful API endpoints for RaceDay prior to the implementation of the application. The design specifies how system entities are related, what data must be stored, and how users and applications can utilize the API.

The Organiser and Participant roles can be authenticated by the system. Protected pages of the system require authentication, and user passwords should be hashed before storing them in the database.

2. On our System Roles we have:

- User
- Event
- Category
- EventCategory
- Enrolment
- Result

User–Event: An Organiser can organise many Events, but each Event is linked to one Organiser. This is a one-to-many relationship. The OrganiserID foreign key in the Event entity refers to the UserID primary key in the User entity.

Event–Category: An Event can belong to many Categories, and a Category can contain many Events. This is a many-to-many relationship. The EventCategory junction entity resolves this relationship by holding the EventID and CategoryID as foreign keys and a composite primary key.

User–Enrolment: A Participant can have many Enrolments, but each Enrolment belongs to one User. The UserID foreign key in Enrolment is a reference to the User entity.

Event–Enrolment: An Event can have many Enrolments, but each Enrolment is related to one Event. The EventID foreign key in Enrolment is a reference to the Event entity.

Enrolment–Result: An Enrolment can have zero or one Result. An enrolment may exist before an event occurs, which means a result is not immediately available. Only after the participant completes the event can their result be entered. The EnrolmentID foreign key in Result refers to the Enrolment entity.

SECTION B: API Endpoint Plan

RaceDay will provide a RESTful API to enable users to interface with the application and have it performed the functions defined by the system Scope of work. The following is a plan of the api endpoints with route, method, brief description, body, and response (role-based required).

The API covers the key functional aspects of RaceDay, such as authentication, user profile, events, categories, enrolling in events and results.

There are some public endpoints that can be accessed without authentication and other protected endpoints that require the user to be logged in. Role based access control is implemented for certain features either for the Organiser or Participant.

Users sign up by providing some information about themselves and selecting a role. Passwords are not stored as plain text. Rather than storing the password in the database, the password has been hashed. Authenticated users can then utilize protected API functionality based on their role.

AUTHENTICATION ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected responses
POST	/api/auth/register	Creates a new RaceDay user with the given personal details, email address, password and role. The password is hashed and saved to the database.	None (Public)	{firstName, lastName, email, password, role}	201 Created – account created successfully. 400 Bad Request – you have an error in your registration, or you did not fill in the form completely. 409 Conflict –

					the email address already exists
POST	/api/auth/login	Verifies a registered user by email and password and returns authentication information for use with protected endpoints.	None (public)	{email, password}	200 OK – login was successful and the authentication token provided. 400 Bad Request – Login details not provided. 401 Unauthorized - Your email or password is wrong.

USER PROFILE ENDPOINTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Answers
GET	/api/profile	Gets the user information of the current user.	Any registered user	None	200 OK – the user profile has been successfully returned. 401 Unauthorized – the user has not been authenticated.
PUT	/api/profile	Updates the information of the user that is currently authenticated.	Any registered user	{firstName, lastName}	200 OK - the profile was updated successfully. 400 Bad Request - profile data is invalid. 401 Unauthorized - the user is unauthenticated.

The profile route does not require a user ID in the URL, as the system knows who the current user is via their authentication information. This stops people from trying to access or modify other people's private profiles.

EVENT ENDPOINTS

HTTP Method	Route	Description	Role required	Request Body	Expected answers
GET	/api/events	Get a list of available RaceDay events.	None (public)	None	200 OK - The list of events is returned successfully
GET	/api/events/{id}	Gets information about a single event by its event ID.	None (public)	None	200 OK – event successfully returned. 404 Not Found – the event does not exist.
POST	/api/events	Creates a new event. The event is owned by the authenticated Organiser.	Organiser	{name, description, eventDate, location, distanceKm, capacity}	201 Created – the event has been successfully created. 400 Bad Request – the event information is invalid or incomplete. 401 Unauthorized – user is not logged in. 403 Forbidden – authenticated user is not an Organiser.
PUT	/api/events/{id}	Modifies organiser details of an event for which the current user is the organiser.	Organiser	{name, description, eventDate, location, distanceKm, capacity}	200 OK – event updated successfully. 400 Bad Request – invalid details for the event. 403 Forbidden – The organiser does not have permission to manage this event. 404 Not Found – the event could not be found
DELETE	/api/events/{id}	Removes an	Organiser	None	204 No

		existing event managed by the authenticated Organiser.			Content— Event deleted successfully 403 Forbidden— The organiser is not allowed to delete the event. 404 Not Found – event with the specified id does not exist.
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CATEGORY ENDPOINTS

HTTP Method	Route	Description	Role required	Request body	Expected answer
GET	/api/categories	Gets all the categories in RaceDay.	None (public)	None	200 OK – categories listed successfully.
GET	/api/events/{eventId}/categories	Returns all categories linked to an event.	None (public)	None	200 OK – event categories have been returned successfully. 404 Not Found – the event does not exist.
POST	/api/categories	Creates a new group that can be linked to RaceDay races.	Organiser	{name, description}	201 Created – category successfully created. 400 Bad Request – category information is invalid. 409 Conflict – A category with the same name already exists.
POST	/api/events/{eventId}/categories/{categoryId}	Links an	Organiser	None	201 Created

	oryId}	existing category to an event.	er		– Category assigned to the event successfully. 403 Forbidden - Organiser does not have permission to manage the event. 404 Not Found – event or category can't be found. 409 Conflict – category already attached to the event.
DELETE	/api/events/{eventId}/categories/{categoryId}	Erases the connection between the category and the event.	Organiser	None	204 No Content – The association between the category and the event has been successfully removed. 403 Forbidden – The organiser is not authorised to manage the event. 404 Not Found – The event/category association does not exist.

EVENT ENROLMENT ENDPOINTS

HTTP Method	Route	Description	Role required	Request Body	Expected answers
POST	/api/events/{eventId}/enrolments	Register the current user as a Participant in a particular event.	Participant	None	201 Created – the enrolment was created successfully. 400 Bad Request – the user cannot be enrolled as a participant, for example if the event has no more seats available or is invalid. 401 Unauthorized – user is not logged in. 403 Forbidden – you are authenticated but not a Participant. 404 Not Found – no such event. 409 Conflict – the participant has already enrolled.
GET	/api/enrolments	Get a list of all the event enrolments for the current user.	Participant	None	200 OK – participant enrolments are successfully returned. 401 Unauthorized – user has not been authenticated. 403 Forbidden – an authenticated user is not able to join as

					a Participant.
GET	/api/events/{eventId}/enrolments	Gets all participant enrolments for a specific event that the Organiser can use to manage the event participants.	Organiser	None	200 OK – event enrolments successfully returned. 403 Forbidden – Organiser is not authorised to manage the event. 404 Not Found – event does not exist.
DELETE	/api/enrolment/{id}	Cancels or deletes an enrolment of the currently authenticated Participant.	Participant	None	204 No Content – The enrolment has been cancelled or deleted successfully. 403 Forbidden – the enrolment is not that of the authenticated Participant. 404 Not Found – the course enrolment does not exist.

The enrolment entity is what links the Participants and Events. A participant can sign up for multiple events, and a single event can be attended by many participants.

RESULT ENDPOINTS

HTTP Method	Route	Description	Role required	Request body	Expected answer
POST	/api/enrolments/{enrolmentId}/result	Logs a result for a Participant who has attended a specific	Organiser	{finishTime, position}	201 Created – the result has been successfully recorded. 400 Bad

		event.			Request – information about the result is invalid. 403 Forbidden – the Organiser is not permitted to manage the event in question. 404 Not Found – the enrolment does not exist. Conflict – a result for the enrolment already exists.
PUT	/api/results/{id}	Modifies an already existing Participant result.	Organiser	{finishTime, position}	200 OK – result successfully updated. 400 Bad Request – the result information is invalid. 403 Forbidden – Organiser is not permitted to manage the result. 404 Not Found – result does not exist.
GET	/api/events/{eventId}/results	Gets the recorded results for a particular RaceDay event.	None (public)	None	200 OK - the event results have been returned successfully. 404 Not Found – the event does not exist.
GET	/api/results/me	Gets the	Participa	None	200 OK – the

		current authenticated Participant's personal performance and result history from past events.	nt		history of the personal result was successfully returned. 401 Unauthorized – authentication of the user has failed. 403 Forbidden – you are authenticated, but not a Participant.
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HTTP Methods Used

HTTP Method	Purpose
GET	Returns information already in the system.
POST	Creates a new resource or record.
PUT	Modify an existing resource or record.
DELETE	Delete an existing resource or record.

Authentication and role-based access

The RaceDay API differentiates between public and protected methods.

Public API endpoints, so things like viewing events, categories and event results, these do not require you to be logged in. Valid authentication is necessary to access protected endpoints.

There are two user roles supported by RaceDay:

Organiser — the person who creates the events, manages the event categories, checks event enrolment and completes or modifies the results.

Participant - can view events, register for events, manage their own registrations, and view their performance history.

The API will check the authenticated user's role prior to providing access to role-restricted endpoints. An authenticated user with no roles assigned is not authorized to use the userInfo endpoint, which would return a 403 Forbidden.

You will receive a 401 Unauthorized response if an unauthenticated user tries to get to a protected resource.

Password and security

At registration, users supply their password in the registration request. The password of the user should not be stored in the database as is.

Pre-saving user account, the app will hash the password by means of a suitable password-hashing function. Just the resulting password hash will be stored in the PasswordHash field of the User entity.

Upon login, the user-provided password is securely compared to the stored password hash. This means that now the credentials are hashed before being stored in the RaceDay DB, and user's credentials are more secure.

API and database alignment

The API endpoint structure is intended to align with the entities from the RaceDay erd.

This alignment allows the API functionality to be implemented using the relational database structure defined in Section A.

For instance, a Participant signing up for an Event via:

POST /api/events/{eventId}/enrolments

the system generates an Enrolment record composed of the Participant's UserID and the chosen EventID.

In the same way, an Organiser adding a category to an Event via:

POST /api/events/{eventId}/categories/{categoryId}

the system generates the appropriate record in the EventCategory junction table.

Conclusion

The RaceDay API endpoint schema defines how the RaceDay clients interact with RaceDay services. The endpoints are enough to implement all the necessary features for login, user profiles, events, categories, event enrolments and results.

The API section describes various HTTP success and failure responses and specifies whether an endpoint is public, for all authenticated users or restricted to a specific role. Authentication and author-based access control are used to encrypt the user credentials so that they can be stored in a secure way instead of plain text.

It is worth noting that this API endpoint plan also relates to the database entities within the RaceDay ERD and will assist in the construction of the RaceDay API in the next development phase.